

PORT OPERATIONS INTELLIGENCE

HARBORLINE

Container Congestion Predictor & Port Operations Optimiser

BugHunters

DEPSTAR, CHARUSAT

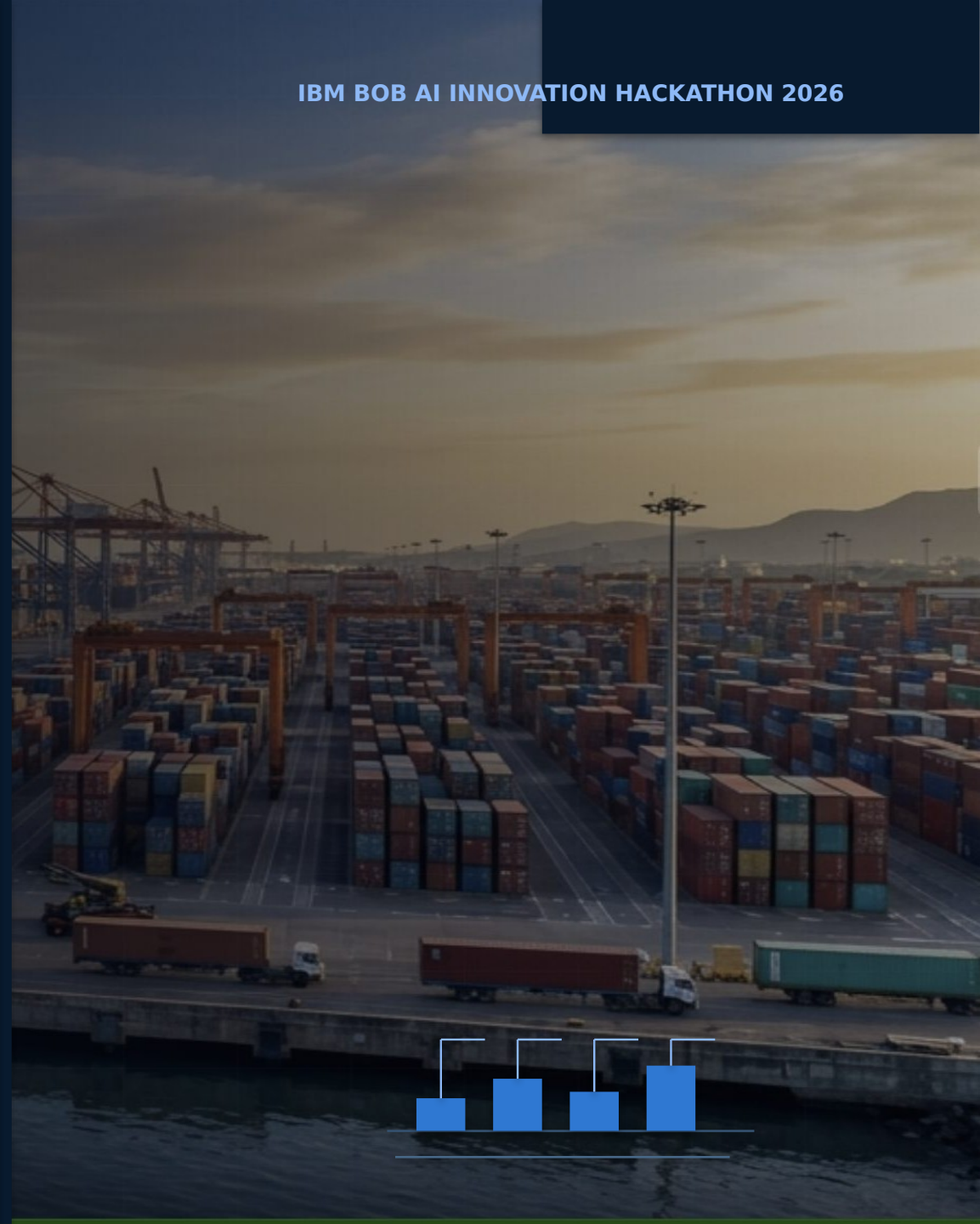
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Indian ports covered

24h · 48h · 72h **XGBoost**

Forecast horizons

ML model



Port Congestion Is Difficult to Anticipate

Port teams need visibility into current vessel activity and the ability to anticipate congestion pressure over the next 24-72 hours so operational decisions can be made earlier.

WHAT

Pressure builds before congestion is visible.

WHO

Supervisors, berth planners and operations teams.

WHY

Earlier awareness supports better sequencing and resource decisions.

855M+

tonnes handled across the operating network

48.84h

average time vessels spend in port

Reactive decisions leave too little time to prevent the queue.

OPERATIONAL CHAIN

Vessel Activity



Increasing Port Pressure



Congestion



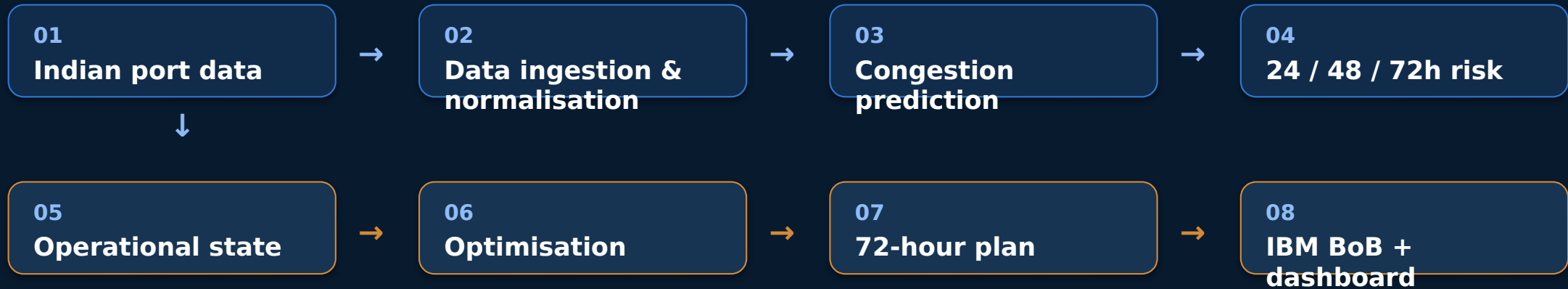
Waiting / Resource Conflicts



Schedule Disruption

Harborline

Harborline combines historical port activity, vessel intelligence and operational information to forecast port-level congestion pressure and connect the forecast with operational optimisation.



ML predicts **port-level congestion pressure**. The optimisation layer handles vessel sequencing, berth allocation and resource constraints.

PREDICT → OPTIMISE → EXPLAIN

Technical Architecture

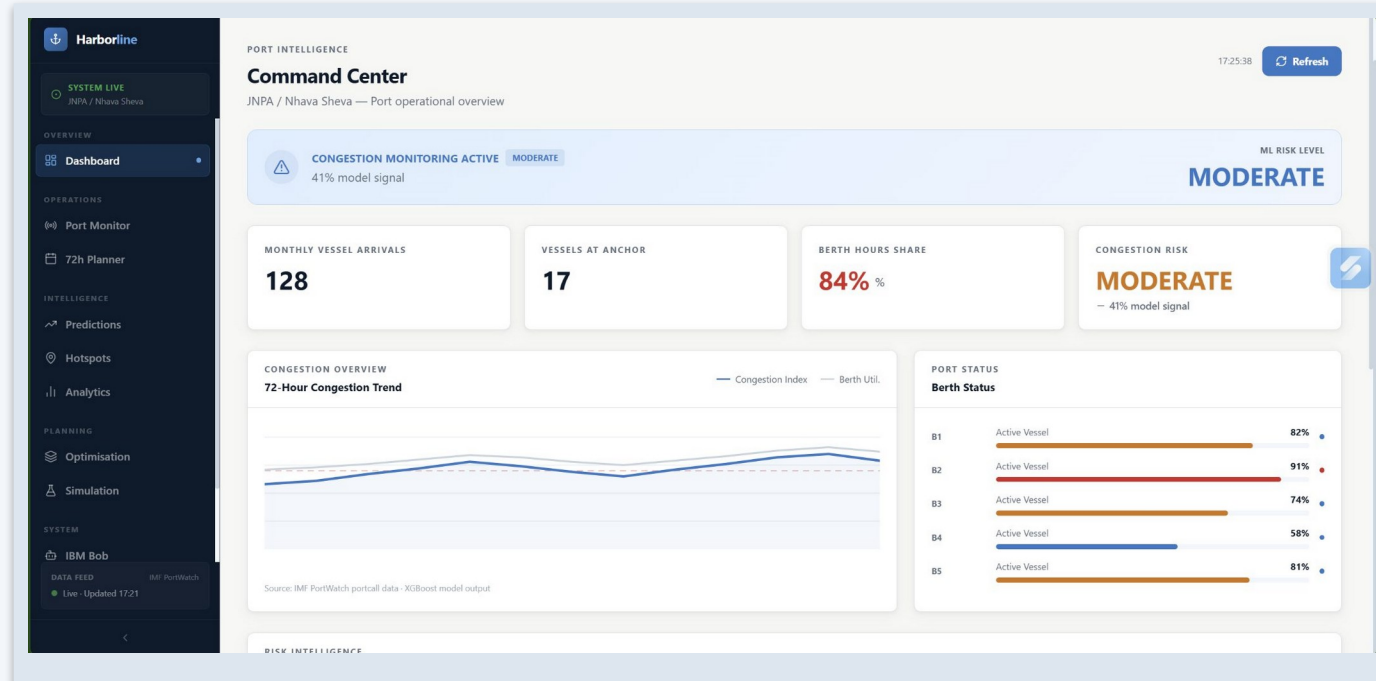


The architecture separates data quality, prediction, operational decisioning and supervisor access so each layer can be validated independently.

INDIAN PORT FOCUS

**Mundra · JNPA · Chennai ·
Kandla · Visakhapatnam**

24-72 Hour Congestion Intelligence



ATTACHED PRODUCT VIEW · JNPA / NHAVA SHEVA

FORECAST HORIZONS

24h → 48h → 72h

Risk score and pressure state over the planning window.

PRESSURE STATE

BUILDING

STABLE

SURGING

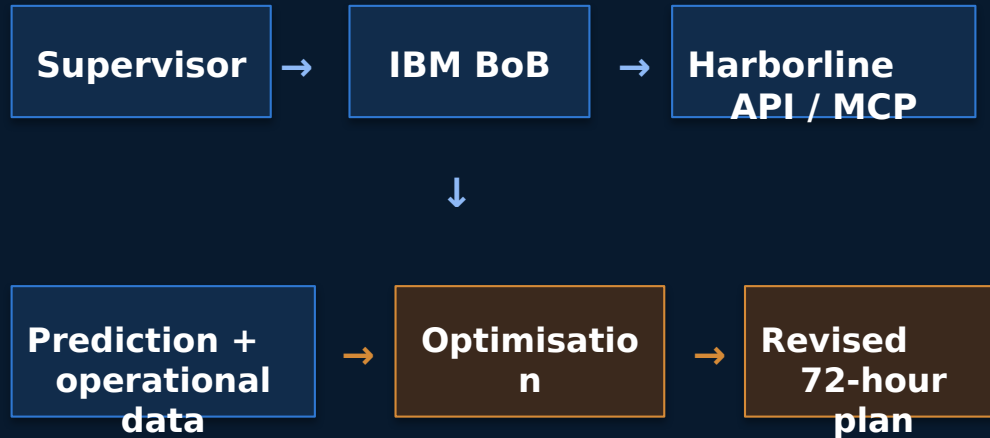
RECOVERING

RISK DRIVERS

Activity momentum
Historical activity baseline
Cargo activity
Short-term volatility

IBM BoB as an Operational Interface

OPERATIONAL WORKFLOW



EXAMPLE INTERACTIONS

Supervisor

“Why is Mundra showing increasing congestion risk?”

BoB Retrieves Harborline data → reads prediction → identifies risk drivers → explains the situation.

Supervisor

“What happens if a vessel is delayed by 6 hours?”

BoB Runs a what-if workflow → invokes optimisation → returns a revised plan.

**Natural-language operations, backed by tools—
not a chatbot wrapper.**

MCP tools: get_port_status · optimise_schedule · generate_72_hour_plan

Results & Engineering Validation

VALIDATED MODEL METRICS

ROC-AUC by forecast horizon

Chronological evaluation of the 24h / 48h / 72h congestion-risk classifier.

24H

0.829

ROC-AUC

48H

0.837

ROC-AUC

72H

0.825

ROC-AUC

Requested metric view: ROC-AUC only.

Evaluation uses chronological train/test separation and point-in-time feature construction.

PRACTICAL IMPACT PATH

Forecast



Earlier awareness



Operational response



Revised plan

72-hour forward visibility shifts decisions from 0-hour reaction to a 3-day planning window.

BugHunters

DEPSTAR, CHARUSAT

FOUR DISCIPLINES · ONE OPERATING PRODUCT

01

Aubaid

Data + AI / ML

Real-data congestion forecasting engine

IMF PortWatch and public port data
Feature engineering + cleaning
XGBoost risk forecasts

02

Abdulkadir

Backend + Optimisation

Decision engine + IBM BoB integration

FastAPI services and PostgreSQL
Berth, vessel and crane optimisation
MCP tools + what-if scenarios

03

Ansh

Frontend + Product

Operational dashboard and congestion views

Command center visualisation
72-hour planning views
What-if UI + presentation

04

Krish

Frontend + Product

Debugger and overall structure

Handled docs and GitHub repo
Bug fixes
Optimised overall structure

HARBORLINE

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